

CoffeeShopDB

SQL Business Analysis — Guided Practice Project

A practical, phased project for developing SQL skills and Business Analyst thinking

Goal: Move from writing SQL queries to answering real business questions with data.

Scope: SQL only for now. Excel and Power BI are intentionally excluded.

How to Use This Guide

For every question, follow the same process:

1	Understand	Rewrite the business question in your own words.
2	Identify	Choose the tables and columns you need.
3	Query	Write the SQL using only concepts you have learned.
4	Validate	Check joins, NULLs, duplicates, filters, and totals.
5	Explain	Explain what the result means in business language.
6	Reflect	Ask what decision the business could make from the result.

Data Quality & Analysis Policy

Use these rules throughout the project so your analysis remains consistent and defensible.

Issue	Rule
NULL	Determine what NULL means before replacing, excluding, or calculating with it.
Missing values	Do not automatically assume missing = 0.
Duplicates	Check whether duplicates are legitimate records or data errors.
Data types	Make sure dates, numbers, and text are stored and treated appropriately.
Dates	Use consistent date definitions; when grouping by month, consider the year as well.
Negative values	Investigate unexpected negative amounts before removing them.
Zero values	Distinguish a genuine zero from a missing value.
Joins	Check whether a join creates duplicate rows or changes totals.
Outliers	Investigate unusual values before deciding how to handle them.
Calculations	Document important assumptions behind calculations.
Filtering	Be clear about which records are included or excluded.
Validation	Validate important results before using them for business conclusions.

Core principle: Never manipulate a data-quality issue simply to make the query produce a cleaner result. Understand the meaning of the data first, document the assumption, then apply the appropriate SQL treatment.

Phase 1 — Business Performance

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: How many orders has the coffee shop received?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: What is the total revenue generated?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: What is the average order value?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: How many orders are completed, pending, or cancelled?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: Which period generated the highest revenue?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Phase 2 — Product Analysis

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: Which products generate the most revenue?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: Which products sell the most units?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: Which products have the highest average selling price?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: Which products have low sales activity?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: How does product performance vary across categories?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Phase 3 — Customer Analysis

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: *How many customers have placed orders?*

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: *Which customers have spent the most?*

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: *Which customers have placed the most orders?*

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: *Which customers have never placed an order?*

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: *How much revenue does each customer contribute?*

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Phase 4 — Time Analysis

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: How does revenue change over time?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: Which month generates the highest revenue?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: How many orders are placed each day or month?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: Are there periods with unusually low or high sales?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: How does current-period performance compare with previous periods?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Phase 5 — Operational Analysis

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: Which suppliers provide the products used by the business?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: Which products or suppliers may require attention based on sales activity?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: Are there products with missing or incomplete information?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: Do joins between operational tables change the expected record counts?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: Which operational records should be investigated for data-quality issues?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Phase 6 — Employee Analysis

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: How many employees are represented in the database?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: Which employees are associated with the most orders or activity?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: How does activity vary between employees?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: Are there employees with no associated records?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: Which employee-related patterns could be useful for management review?

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Phase 7 — Advanced Analysis: Window Functions

Do not jump straight to the SQL. Start by identifying the business meaning, tables, grain of the data, and required calculation.

Question 1: Rank products by revenue.

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 2: Rank customers by total spending.

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 3: Calculate a running total of revenue over time.

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 4: Compare each period's revenue with the previous period.

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

Question 5: Calculate each customer's contribution as a percentage of total revenue.

Tables/columns: _____

SQL concepts: _____

Business interpretation: _____

SQL Concepts Checklist

■	SELECT / FROM
■	WHERE and conditions
■	ORDER BY
■	JOINS
■	Anti-joins
■	GROUP BY
■	Aggregation
■	CASE
■	NULL handling
■	Date & time functions
■	Set operators
■	Window functions
■	Data validation
■	Business interpretation

Business Analyst Reflection

After completing a question, write one or two sentences answering:

- What did the query tell me?
- Why would the business care about this result?
- What action or decision could this information support?
- What assumptions or data-quality issues could affect the result?

Completion Standard

The goal is not simply to finish all 20 questions. A question is considered complete when you can write the SQL, explain why it works, validate the result, and explain its business meaning without relying on the query itself.